

7 Electronic Construction

7.1 System Overview

The electronic system of Glidesat consists of three custom-designed printed circuit boards (PCBs): a mainboard, a motor control board, and a Linux board. Off-the-shelf solutions were evaluated but rejected because their physical dimensions were not suitable for the 66mm diameter CanSat form factor and they provided no isolation between high-current actuator paths and sensitive sensor/communication circuitry. Splitting the electronics across three boards allowed the us to fit all required functionality within the diameter limit while achieving electrical isolation between subsystems.³

7.1.1 Architecture

- **Mainboard** – flight computer, sensor, LoRa telemetry, GPS, data logging, power distribution.
- **Motor control board** – parafoil actuator drive, burn wire mechanism, buzzer recovery system.
- **Linux board** – Linux SBC, high-resolution imaging, onboard image storage.

This separation isolates the high-voltage, high-noise actuator paths on the motor board from the sensitive GPS, IMU, and LoRa circuitry on the mainboard. Each board operates on its own regulated power rail, providing independent fault domains.

7.1.2 Block Diagram

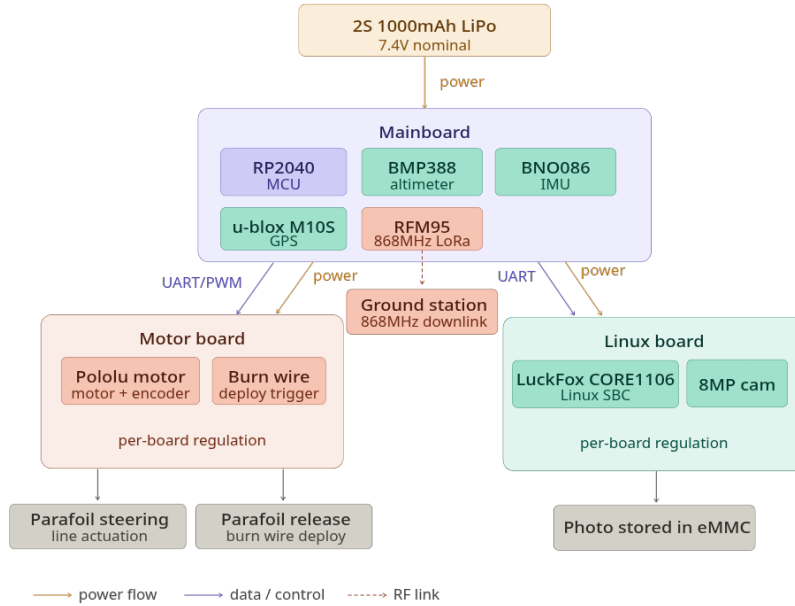


Figure 16: Block Diagram

7.2 PCB Design

7.2.1 Board Specifications

All boards are circular with a diameter of 60 mm and use M2.5 mounting holes positioned 15 mm from the centre of the board. The mainboard and Linux board use a four-layer stackup with the following layer assignment:

- L1 – signal

- L2 – ground plane
- L3 – power (with a dedicated region for raw battery voltage on the mainboard to improve current handling)
- L4 – signal

The motor board uses a two-layer stackup sufficient for its lower component density.

7.2.2 PCB Images

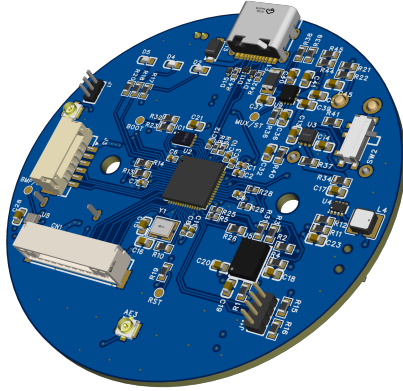


Figure 17: Mainboard



Figure 18: Mainboard(bottom)

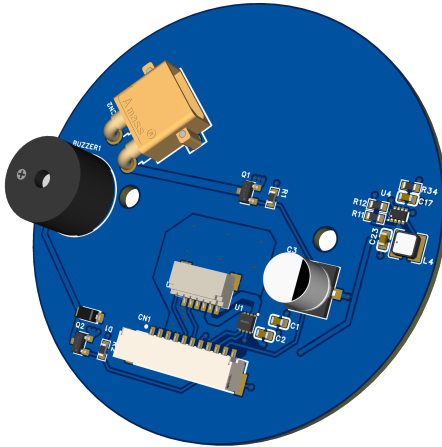


Figure 19: Motor Board

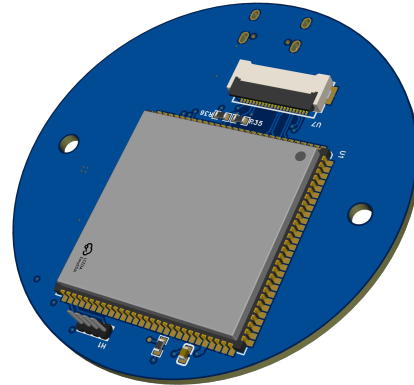


Figure 20: Linux Board

7.2.3 Board-to-Board Connections

JST-GH 1.25 mm pitch connectors were selected for all inter-board signal connections. The positive locking mechanism prevents disconnection under vibration during rocket ascent and mechanical shock on landing – a failure mode that unlatched connectors such as JST-PH or Dupont headers are susceptible to. All cables include strain relief at both ends.

7.3 Subsystem Descriptions

7.3.1 Flight Computer – RP2040

The RP2040 dual-core Cortex-M0+ SoC act as the flight computer.

Peripheral connections: SPI for RFM95W LoRa module and W25Q16JV flash memory; I2C for BMP388 and BNO086; UART for u-blox MAX-M10S and LuckFox Core1106; PWM for DRV8838 motor driver; GPIO ADC for battery voltage monitoring. A 12MHz crystal oscillator provides the system clock reference.

7.3.2 Sensor Subsystem

Table 6: Sensor subsystem summary

Component	Function	Interface	Key Spec
BMP388	Barometer, temperature	SPI/I2C	± 0.5 m altitude resolution
BNO086	9-DOF IMU	I2C	On-chip sensor fusion, quaternion output
u-blox M10S	GPS / GNSS	UART	Concurrent GNSS, 9 mA typical
IMX415	Camera sensor	MIPI CSI	8 MP, via LuckFox Core1106

BMP388 The BMP388 provides barometric altitude and temperature measurements. Its altitude resolution of ± 0.5 m and low noise floor are important for monitoring the gliding descent without camera assistance. The higher sampling rate of up to 200 Hz and improved temperature compensation over its alternatives make it well suited to the Glidesat body that oscillates during descent.

BNO086 A 9-DOF IMU with on-chip sensor fusion was required because a 6-DOF IMU without a magnetometer causes yaw drift, which leads to mis-steering of the parafoil. The BNO086 outputs calibrated data directly, removing the need for an external fusion algorithm, reducing MCU load, and removing the risk of incorrect manual calibration.

u-blox MAX-M10S The MAX-M10S supports concurrent reception of GPS, GLONASS, Galileo, and BeiDou signals, improving position accuracy and fix reliability. At approximately 9 mA typical current consumption it draws significantly less power than the Neo-6M module used in earlier revisions. A 0.33 F supercapacitor on the mainboard maintains the module’s hot-start data and real-time clock during brief power interruptions, reducing position reacquisition time from approximately 30 s (cold start) to approximately 1 s (hot start).

7.3.3 Communication Subsystem

The RFM95W-868S2 LoRa transceiver operates at 868 MHz, which is within the licence-free ISM band across the European Union. The module is connected via SPI and implements a bi-directional link:

- **Downlink (CanSat → ground station):** altitude, temperature, GPS coordinates, battery voltage, system status, camera capture status.
- **Uplink (ground station → CanSat):** motor direction, motor speed, recovery system activation.

All telemetry is simultaneously logged to the W25Q16JV 16 Mbit flash memory, providing a complete flight data record independent of the radio link.

Antenna note: During PCB layout review it was identified that impedance matching for the RFM95W antenna trace was not implemented and the copper ground plane was not cleared under the antenna connector. This is expected to introduce approximately 1–3 dB of additional insertion loss

due to signal reflection. The ground station uses a 13 dBi Yagi antenna, which provides sufficient gain margin to compensate for this loss. The link was confirmed operational during range testing. Also, a empty pi network with 0Ω resistor was implemented for post-impedance matching. So we can tune the impedance if we need extra range.

7.3.4 Actuation Subsystem

The motor board drives a Pololu 100:1 micro metal gearmotor with encoder using a DRV8838 single H-bridge motor driver. The 100:1 gear ratio was selected to provide sufficient torque to pull parafoil control lines under canopy load at operational descent speeds. The encoder provides closed-loop position feedback to the RP2040 control loop. The DRV8838 supports up to 1.8 A peak current, which is sufficient for the selected motor, and its WSON-8 2×2 mm package contributes to the compact board layout.

The burn wire deployment mechanism is triggered by a pair of AO3400A N-channel MOSFETs driven directly from RP2040 GPIO. The MOSFET pair allows reliable high-current switching at logic-level gate voltage without the bulk and mechanical wear of a relay. Deployment confirmation is provided by monitoring current draw or encoder movement after trigger.

A buzzer is included on the motor board for post-landing recovery assistance. It is placed on the motor board because the higher supply voltage already present on that board is required for audible output level.

7.3.5 Imaging Subsystem

The LuckFox Core1106 runs a Linux operating system and interfaces with the IMX415 8 MP camera sensor via a two-lane MIPI CSI connection. Images are stored to the onboard eMMC. The LuckFox receives image capture commands from the RP2040 over UART and responds with capture status. Because Linux boot time is approximately 20–40 s, the Linux board is powered on and verified before launch to avoid mid-flight boot delay.

7.3.6 Power Subsystem

The system is powered by a 2S 1000 mAh LiPo battery with a nominal voltage of 7.4 V. A TPS2121 power multiplexer on the mainboard allows automatic switchover between battery and USB-C supply, enabling firmware development and sensor testing without connecting the flight battery.

Each board has a dedicated TPS629206 synchronous buck regulator providing the following voltage rails:

- Mainboard: 3.3 V for RP2040, sensors, and LoRa module.
- Motor board: 7.4 V raw battery voltage for motor and burn wire; 3.3 V regulated for logic.
- Linux board: 5 V for LuckFox Core1106.

Per-board regulation ensures that motor switching noise and current spikes on the motor board cannot affect the sensitive GPS, IMU, and LoRa rails on the mainboard. Battery voltage is monitored continuously by the RP2040 ADC using voltage divider. The XT30 battery connector is rated for 30 A continuous, providing a large margin over the 1.3 A peak system current draw.

7.3.7 Protective and Reliability Features

Several protective measures were implemented to improve robustness under the mechanical and electrical stresses of the mission environment.

Table 7: Protective and reliability features

Feature	Component	Protection provided
Overcurrent fuse	1206T4A63V	Trips at 4 A on main power input; protects against wiring faults and short circuits.
ESD protection	USBLC6-2SC6 diode	TVS Clamps electrostatic discharge on USB lines during ground handling and programming.
GPS backup power	0.33 F supercapacitor	Maintains MAX-M10S hot-start data and RTC during brief power interruptions; reduces reacquisition time from ~ 30 s to ~ 1 s.
Locking connectors	JST-GH 1.25 mm	Positive locking prevents vibration-induced disconnection during ascent and landing impact.
Battery connector margin	XT30	30 A rating versus 1.3 A actual peak; eliminates contact resistance and heating concern.
Per-board power isolation	TPS629206 per board	Motor switching noise and fault conditions are contained to the motor board and cannot affect sensor or comms rails.
Decoupling capacitors	100 nF / $1\ \mu\text{F}$ per IC	Bulk and bypass capacitors on every IC supply pin suppress transient voltage spikes.

7.4 Component Selection

7.4.1 Selection Criteria and Weighting

All key component selections were evaluated against seven criteria using a weighted scoring matrix. The criteria and their assigned weights are listed in Table 8.

Table 8: Selection criteria and weights

Criterion	Weight	Rationale
Reliability	3	Flight-critical; no recovery if electronics fail mid-descent.
Performance	3	Directly affects mission success (navigation accuracy, link range).
Power consumption	2	Limited battery capacity; sleep modes required for 4 h operation.
Size	2	60 mm board diameter constraint.
Ease of integration	2	Limited development time.
Cost	1	Secondary to reliability for this application.
Availability	1	All parts can be sourced from various sellers.

Scores from 1 to 5 were assigned for each criterion and each option. The weighted total is the sum of $\text{score} \times \text{weight}$ across all criteria. The chosen part corresponds to the highest weighted total in

every case.

7.4.2 Shortened Decision Table

Table 9: Electronics component selection — Mainboard

Component	Chosen part	Alternative(s)	Reasoning
MCU	RP2040	STM32, ATmega	Dual Cortex-M0+ cores, large community, cheap, PIO for custom peripherals
GPS	u-blox M10S	MAX-Neo-M8N, BN-880	Low power (9 mA), compact package, concurrent GNSS, proven reliability
IMU	BNO086	BNO085, MPU-6050, ICM-42688	On-chip sensor fusion (no extra MCU load), low idle power, interactive calibration
Barometer	BMP388	MS5611, BMP280	± 0.5 m resolution, lower noise floor, low power
LoRa	RFM95W-868S2	SX1276, NRF24L01	868 MHz legal in EU, long range, module reduces RF layout complexity
Flash	W25Q16JV	—	16 Mbit external flash for telemetry and code; RP2040 lacks internal flash
Power MUX	TPS2121	Diode, manual switch	Automatic mux between battery and USB; allows development without battery
Supercap	0.33 F	—	Enables M10S hot start, reducing position acquisition time
Load switch	TPS22810	MOSFET, bulky switch	Limits excessive current; allows smaller switch with low current rating

Table 10: Electronics component selection — Motor Board, Linux Board, and General

Component	Chosen part	Alternative(s)	Reasoning
<i>Motor Board</i>			
Motor driver	DRV8838	L298N, TB6612	Single H-bridge in WSON-8 package; 1.8 A peak, sufficient for our motor
N-MOSFET	AO3400A $\times 2$	Relay, BJT	Reliable high-current switching for buzzer and burn wire
<i>Linux Board</i>			
Linux SBC	Luckfox Core 1106	RPI Zero 2W, ESP32-S3	Built-in MIPI CSI for camera; smaller and lower power than Pi Zero
<i>General</i>			
ESD protection	USBLC6-2SC6 Diode	TVS —	Prevents electrostatic discharge damage during handling
Buck regulator	TPS629206	LM1117, AMS1117	Higher efficiency than linear regulators
B2B connector	JST-GH	JST-PH, SH, etc.	Locking mechanism, reliability
Battery conn.	XT30	JST, screw terminal	Rated for 30 A, massive safety margin, reliable

7.5 Power Budget

7.5.1 Current Consumption

Table 11 lists the estimated current draw for each component in active and sleep states. All values are datasheet typical figures unless noted as measured.

Table 11: Current consumption per component

Board	Component	Active (mA)	Sleep (mA)
5*Mainboard – 3.3 V	RP2040	25	2
	BMP388	3	0.1
	BNO086	6	1
	u-blox M10S	25	15
	RFM95W	30	0.2
2*Motor board – 7.4 V	DC motor	150	5
	Burn wire	0	0
2*Linux board – 5 V	LuckFox Core1106	400	10
	Camera (IMX415)	50	0
Regulators (all boards)	3× TPS629206	30	30
Total		719	63

7.5.2 Mission Profile Calculation

The mission is modelled as 30 min active followed by 3 h 30 min in sleep state. The active duration is a conservative worst-case estimate; typical active time is expected to be shorter.

$$Q_{\text{active}} = 0.5 \text{ h} \times 719 \text{ mA} = 359.5 \text{ mAh}$$

$$Q_{\text{sleep}} = 3.5 \text{ h} \times 63 \text{ mA} = 220.5 \text{ mAh}$$

$$Q_{\text{total}} = 359.5 + 220.5 = 580 \text{ mAh}$$

Assuming a depth of discharge of 70 % to preserve battery longevity:

$$Q_{\text{usable}} = 1000 \text{ mAh} \times 0.70 = 700 \text{ mAh}$$

$$Q_{\text{margin}} = 700 - 580 = 120 \text{ mAh} \quad (17\%)$$

The 120 mAh safety margin is considered sufficient for the 4 h mission requirement.

7.5.3 Battery Selection

The Spektrum Smart G2 2S 1000mAh 50C LiPo was selected. The reduction from the 2700mAh specified at CDR was made possible by the sleep/active power analysis and change in cell number from 1 to 2, which showed that the LuckFox and motor consuming the majority of current are idle for most of the mission. The 50C discharge rating supports a peak current of 50 A, exceeding the 1.3 A actual peak draw (1.3C). The physical dimensions of the pack are compatible with the CanSat body.

7.6 Design Evolution and Revision History

7.6.1 Changes from PDR and CDR

Table 12 summarises all significant design changes made since the PDR and CDR submissions.

Table 12: Design evolution summary

Item	PDR/CDR	FDR	Reason
GPS location	Separate board	Mainboard	Mainboard placement keeps Linux board at bottom; GPS antenna faces top. LoRa/GPS interference negligible at achievable separation.
Camera SBC	RPi CM4 → LuckFox Pico Mini	LuckFox Core1106	CM4 too large/heavy; Pico Mini lacked computing power; Core1106 adds eMMC and robust form factor.
Camera sensor	Arducam 477P → MIS5001 5 MP	IMX415 8 MP	Higher resolution improves landing zone documentation.
IMU	MPU-6050 (6-DOF)	BNO086 (9-DOF)	Magnetometer prevents yaw drift; on-chip fusion reduces MCU load.
Barometer	BMP280	BMP388	± 0.5 m vs ± 1 m resolution; lower noise; higher sample rate.
GPS module	Neo-6M	u-blox MAX-M10S	Concurrent GNSS; 9 mA vs 45 mA; faster fix time.
Motor driver	DRV8833	DRV8838	Single H-bridge sufficient; smaller WSON-8 package.
Battery	1S 2700 mAh (CDR)	2S 1000 mAh	Sleep/active analysis confirms sufficiency with 120 mAh margin; significant weight saving.
Board count	4 boards (separate GPS)	3 custom + 1 commercial	GPS integrated onto mainboard; camera module is commercial.

7.6.2 Revision 1 – Objectives and Findings

The objectives of Revision 1 were to validate the fundamental PCB design: RP2040 bring-up in multi-sensor configuration, SPI / I2C / UART interface verification, power regulation testing, and LoRa radio operation.

7.6.3 Revision 2 – Improvements

Revision 2 implements all lessons from Rev1 testing and upgrades all sensor components to their final flight versions.

Table 13: Test summary

Test	What was verified	Revision	Status
RP2040 bring-up	MCU boot, peripheral init	Rev1	Pass
Sensor communication	I2C/SPI responses	Rev1	Pass
LoRa link	Range, RSSI, packet loss	Rev1	Pass
GPS fix	Fix time, accuracy	Rev2	Pass
Motor + encoder	Closed-loop operation	Rev2	Pass
Burn wire trigger	Fire time, current	Rev2	Pass
Power measurement	Current draw vs estimate	Rev2	Pass
Full system integration	3-board combined test	Rev2	Pass

7.7 Testing and Validation

7.7.1 Test Summary

7.8 Risk Assessment

7.8.1 Risk Table

Table 14: Electronics risk assessment

Risk	Likelihood	Impact	Mitigation
Motor board noise affecting sensors	Medium	High	Per-board power regulation and ground plane isolation.
Burn wire fails to trigger	Low	High	Bench-tested before flight; firmware watchdog triggers backup after timeout if descent rate does not decrease.
LoRa link lost during descent	Low	High	868 MHz chosen for range; all telemetry logged to onboard flash memory independently of radio link.
LuckFox fails to boot	Low	Medium	All flight-critical functions reside on RP2040; LuckFox failure does not abort the mission.
Battery undervoltage during peak draw	Low	High	Decoupling capacitors on each board; motor stall protection in firmware; 120 mAh safety margin.
Connector disconnection on landing	Low	High	JST-GH locking connectors; XT30 battery connector; strain relief on all inter-board cables.
ESD damage during handling	Medium	Medium	USBLC6-2SC6 TVS diode on USB lines; ESD handling procedures during integration.
Antenna impedance mismatch	High	Low	Ground plane not cleared under antenna connector; estimated 1–3 dB loss compensated by 13 dBi Yagi at ground station; confirmed operational in range test.

7.8.2 Residual Risks

The antenna impedance mismatch remains unresolved in Revision 2. This is considered acceptable based on the range test results demonstrating reliable link operation, and the availability of a high-gain ground station antenna. Should field testing reveal insufficient range, the issue will be corrected with empty pi network.